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The Power Behind Mobile Business



The Wealth of Opportunities in the Financial Services Industry

Randy Dence
w-Technologies, Inc.
VP of Business Planning & Development

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About w-Technologies

- Founded in 1997, first to market—first in production (Feb. 1998)
- We can move any type of business into the wireless world
- Rapid development, integration and implementation
- Wireless enterprise architecture can support an entire organization's wireless solution(s)

Wireless Applications in the Financial Services Industry

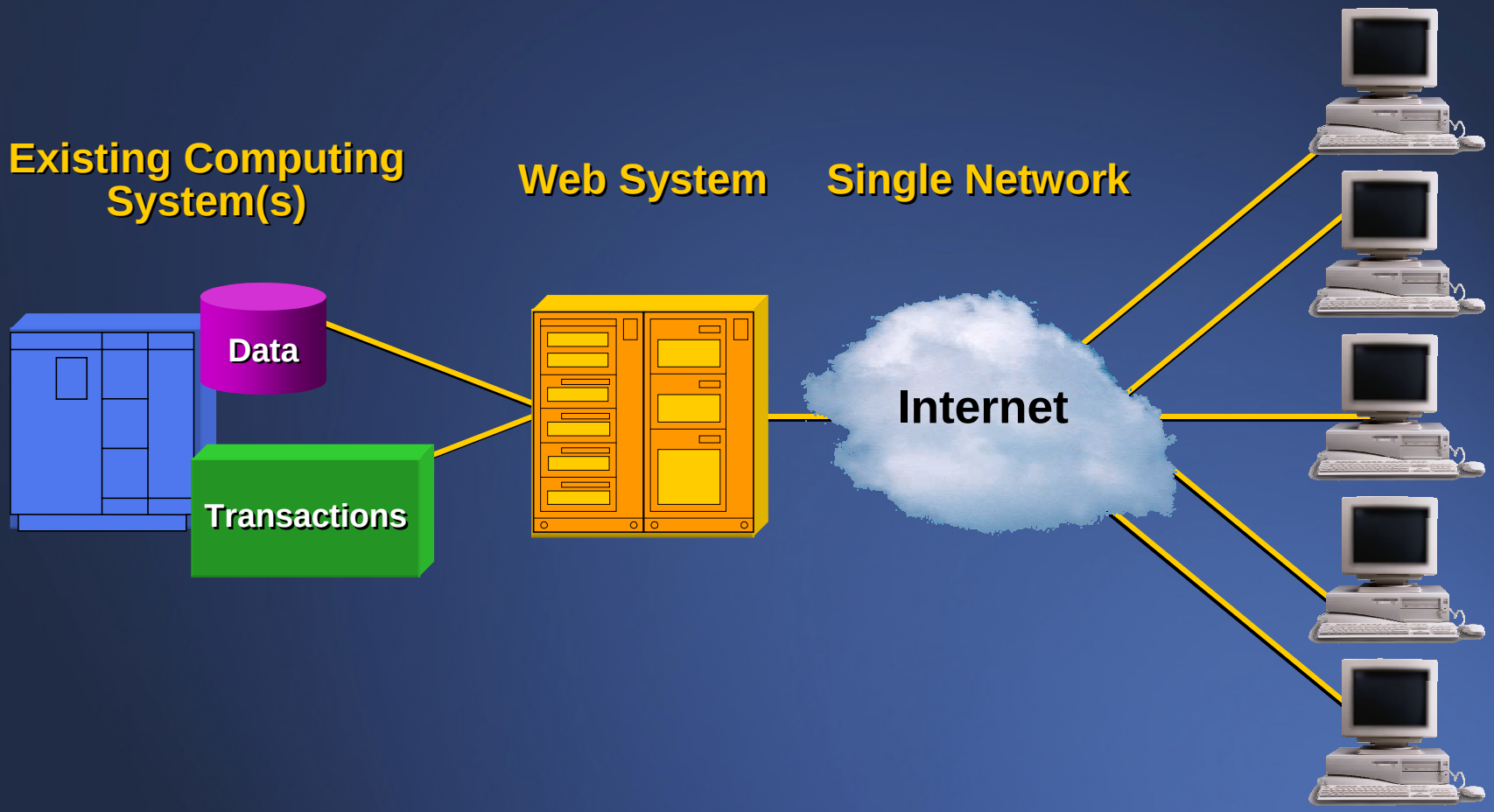
- Wireless becomes a competitive necessity
- Continued movement toward retail financial “portals”
- Use of wireless systems throughout the organization
- Need for a firm wide wireless enterprise architecture

Issues in Developing Wireless Financial Solutions

- Overcoming the “Web-to-Wireless” myth
- Supporting multiple network standards and devices with new ones emerging almost daily
- Flexibility to accommodate evolving business models
- Retaining ownership of the customer/ confidentiality

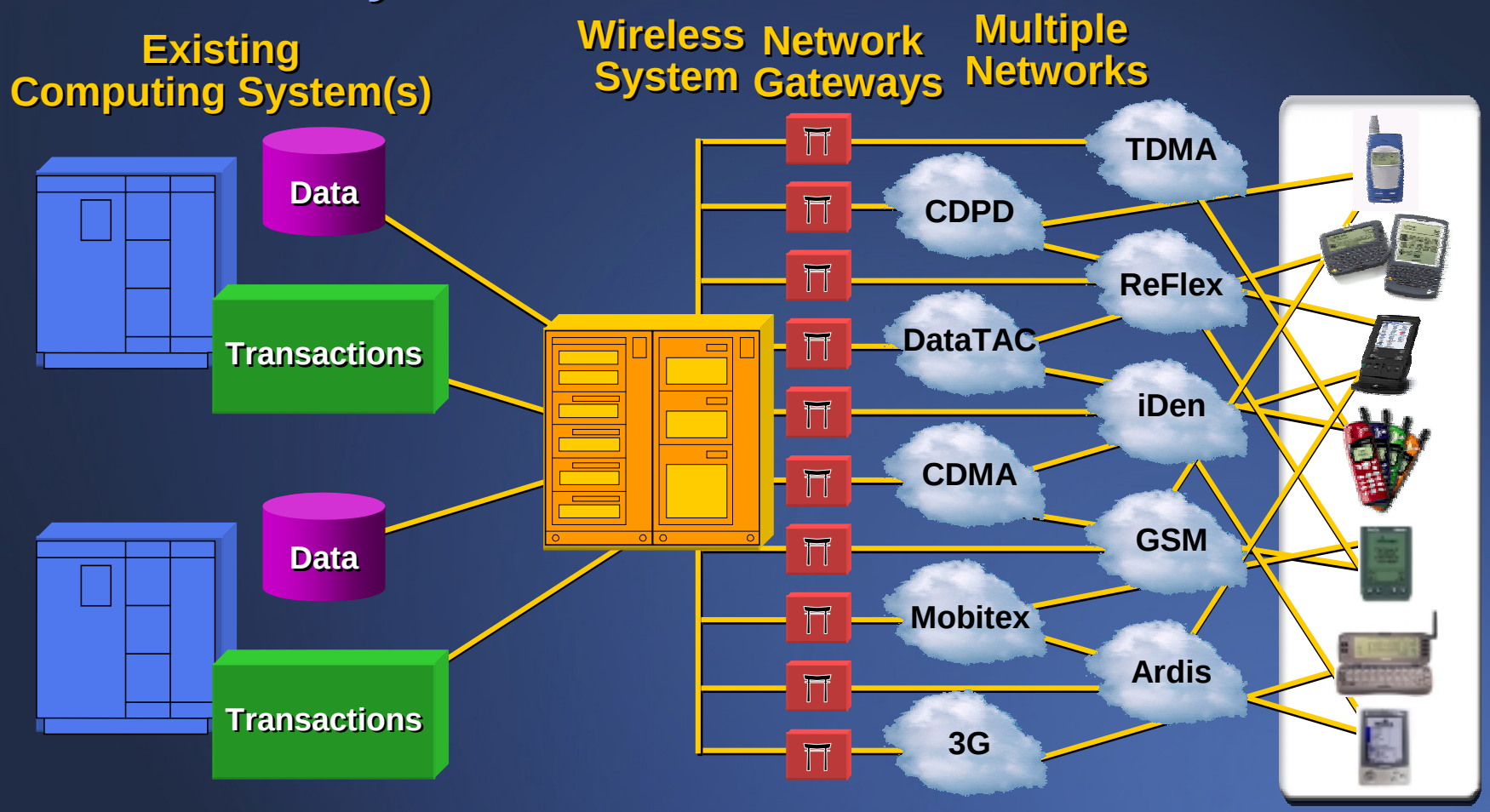
Building the Right Wireless System

Web System Architecture Analogy



Building the Right Wireless System

Wireless System Architecture



Palm.Net® Architecture





BSWD
Mobitex

Internet

Palm.Net

w-Technologies Gateway

Content Feed:
- News
- Quotes
- Sports
- Other

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Application
Server

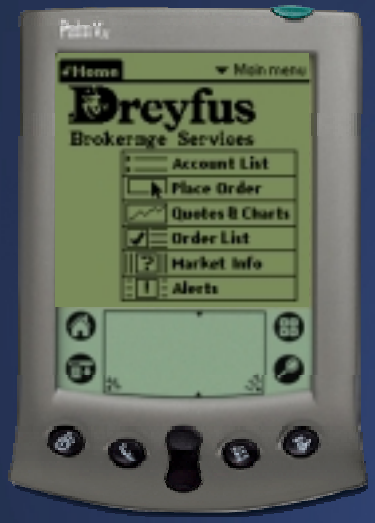
Bank



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Building a Wireless Site That Matches the Business

Dreyfus Brokerage



- **Features/Functions**
  - Trading
  - Real-time Quotes
  - Alerts
  - Standard Trading Condition Features
  - Customized Options Trading Features

SURETRADE

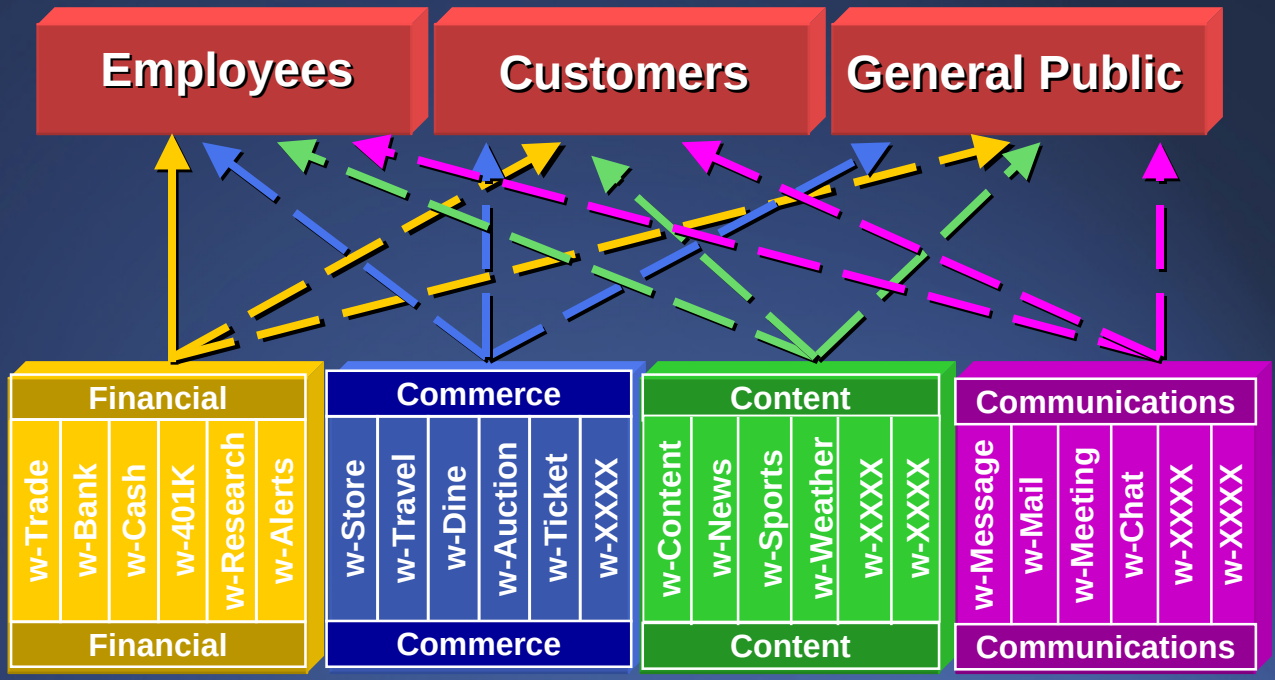


- **Features/Functions**
  - Trading
  - Real-time Quotes
  - Execution Screen to View Trades
  - Alerts
  - Customized Trading Condition Features
  - Standard Options Trading Features

Creating an Extended Mobile Financial Solution

Merrill Lynch





- **Features/Functions**
  - Watch List
  - Real-time Quotes
  - News & Research
  - Office/Seminar Locator
  - Purchase News/Research

The Wireless Future

- “One Standard” will remain elusive
- Convergent devices with added features and functions
- Broadband and 3G
- Location-based services
- Voice capabilities
- Bluetooth
- Wireless “Wallets”

Working Together

- Development Opportunities
- Marketing/Business Development
- Alliances/Mobile Vision Partner (MVP) Program—membership networking





Randy Dence
Vice President
Business Planning & Development

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Security in the Finance World

Gordon Clyne
Security Evangelist

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A Historical Perspective

- Cash is now a financial instrument in a token
- Customers used to carry cash
- Customers are now a walking bubble of digital complexity and trouble!









Challenges to Digital Financial Security

- We are dealing with humans
- Software protocols are complex and not too often standardized around the world
- Some dominant standards are non-U.S. based
- Palm-based solutions NEED to integrate with existing systems
- Redesigning the wheel just for Palm OS® handhelds will fail

PINs and CINs

- Banks refer to their customers by PINs and CINs
- Banks protect this information fiercely
- Banks require end-to-end encryption
- Banks are fudiciarily responsible for your money
- Banks have HUGE arcane systems
- Banks are modernizing FAST
- Banks have big development budgets

Palm Banking From the Customer's Perspective

- If my banking details are in my handheld I better not lose it…
- This is no different than a wallet, perhaps even safer
- Swiping a handheld has a high mental barrier
- Beaming money hasn't really caught on
- Does digital money go away if I lose my data, batteries go flat, etc.

From the Bank's Perspective

- Digital is easier to manage
- Higher profit margins
- Customers have more control of their finances
- More marketing opportunities
- Loyalty programs
- Bank-supplied devices…

Some Future Ideas

- Credentials kept on a server, handheld devices just provide a mechanism to release a PIN
- Magnetic stripe data made beamable
- Re-tooling of every POS terminal to accept IR
- Credentials transmitted via Bluetooth securely
- Smart card readers attached to handheld devices

Finally

- This is a VAST opportunity for application development
- Adequate security services exist today but require licensing
- Basic encryption services will be in the OS less than a year from now


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The Wealth of Opportunities in the Financial Services Industry

Dennis J. Burns
Director, Business Development
Wireless Enterprise Solutions

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Agenda

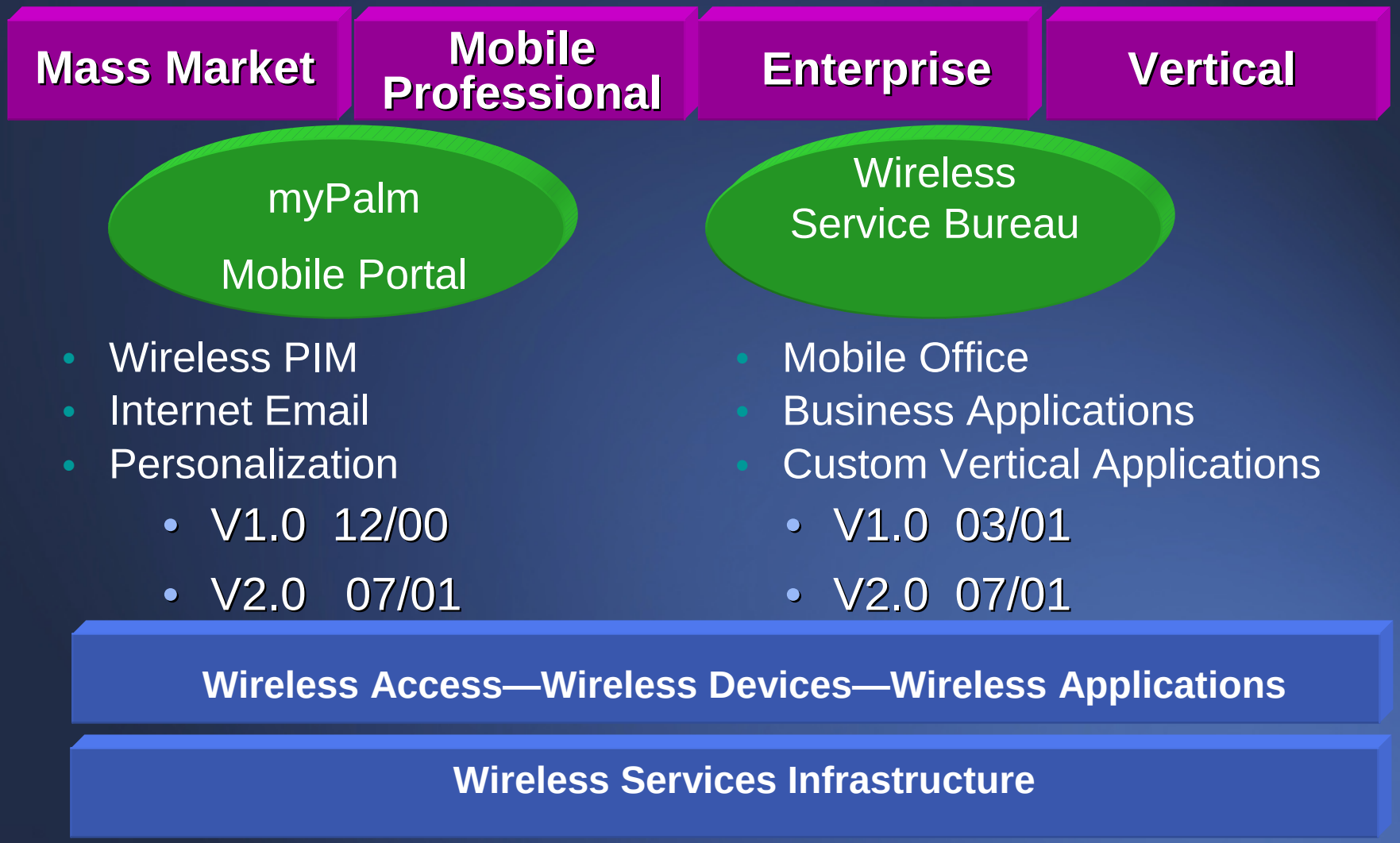
- Content and Access Mission
- Content and Access Strategic Initiatives
- Wireless Enterprise and Access Focus
- Wireless Enterprise Solutions
- Financial Services/Mobile Commerce Trends
  - Drivers
  - Devices
  - Opportunities
- Financial Services / Mobile Commerce
  - Customers and Partners
  - New Partner Products and Services
- Summary
  - What should I do?
  - Palm's Focus on Financial Services

Mission: Leader in Mobile Information Management



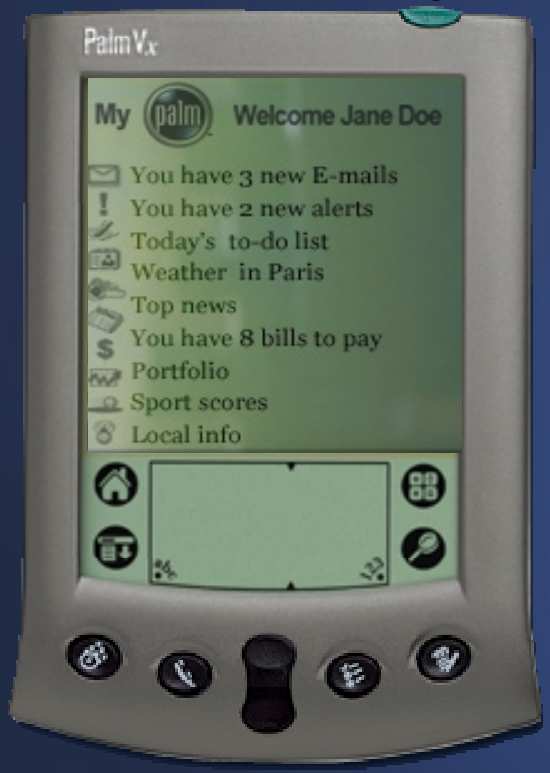
- Extend Palm's leadership as the premier mobile information management tool to being the leading personal and business information management service platform and application provider
  - Always Connected
  - Simple Solutions
  - Compelling Applications

C & A Strategic Initiatives



Palm PIM Portal—
A Seamless Experience

- Personalized experience
- Wireless updating of information



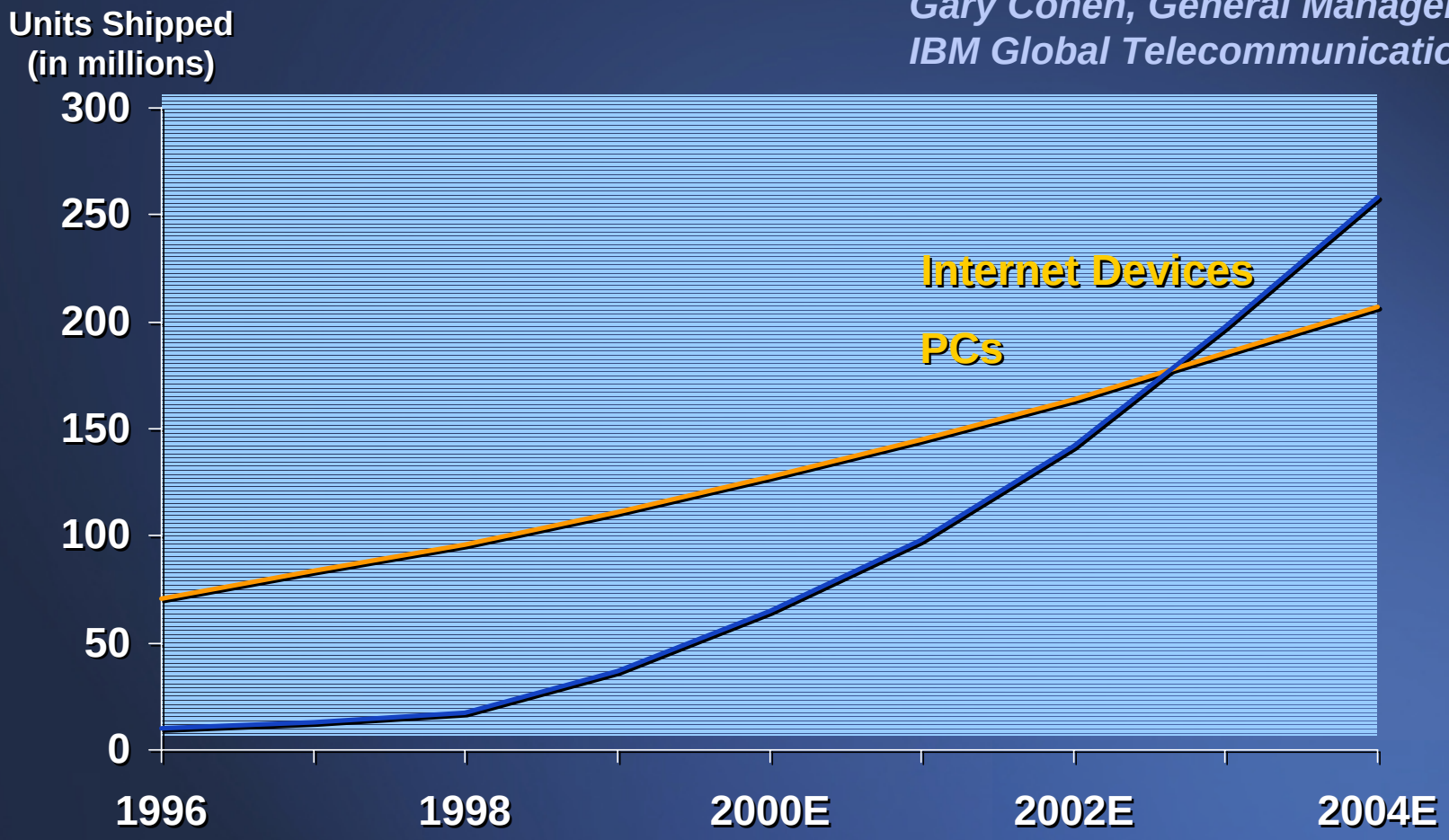


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Devices and Content Are Shifting Away From the PC

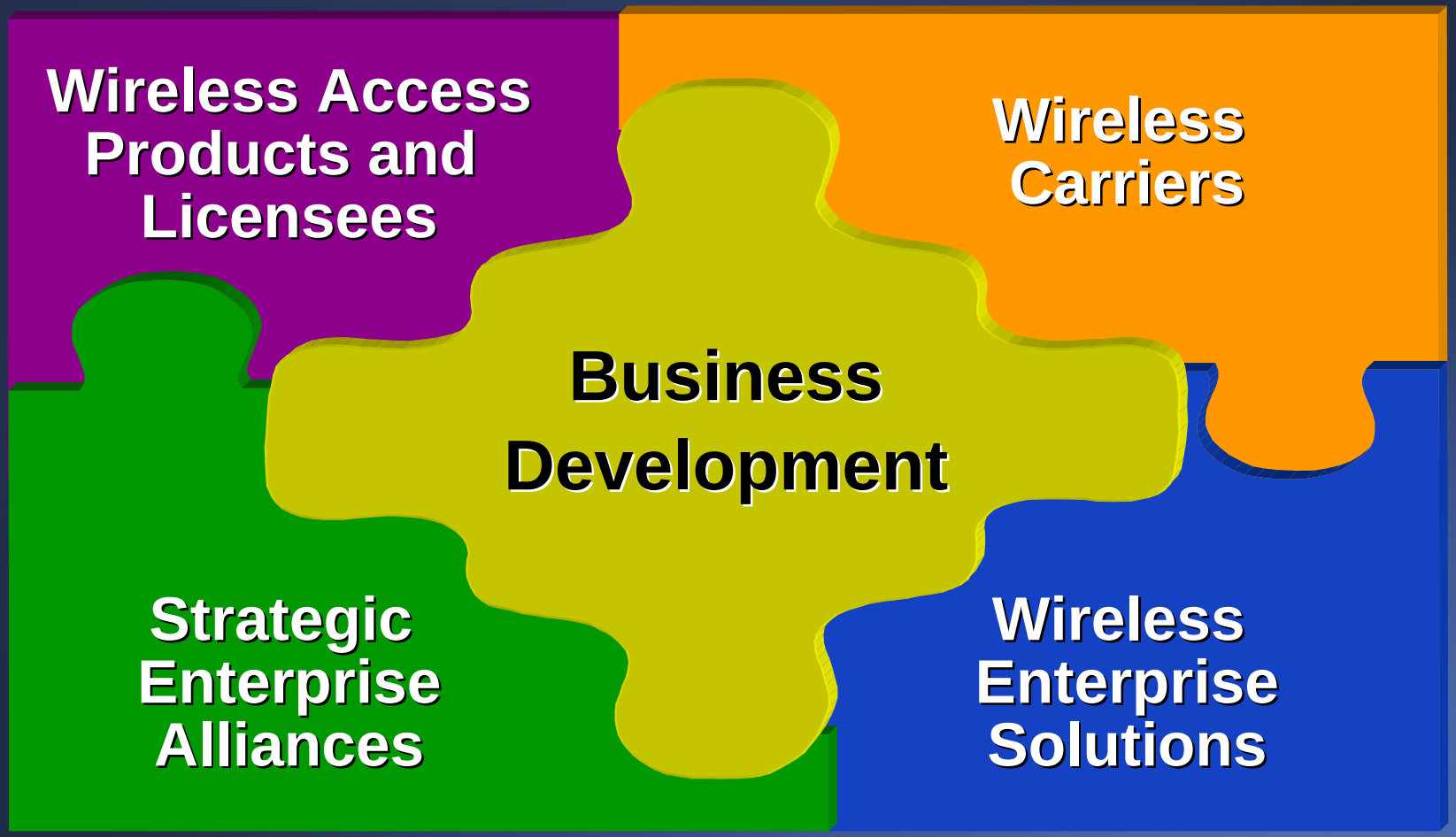
"Over the next five years, more than 80% of new corporate applications will be designed for non-PC devices…"

Gary Cohen, General Manager
IBM Global Telecommunications

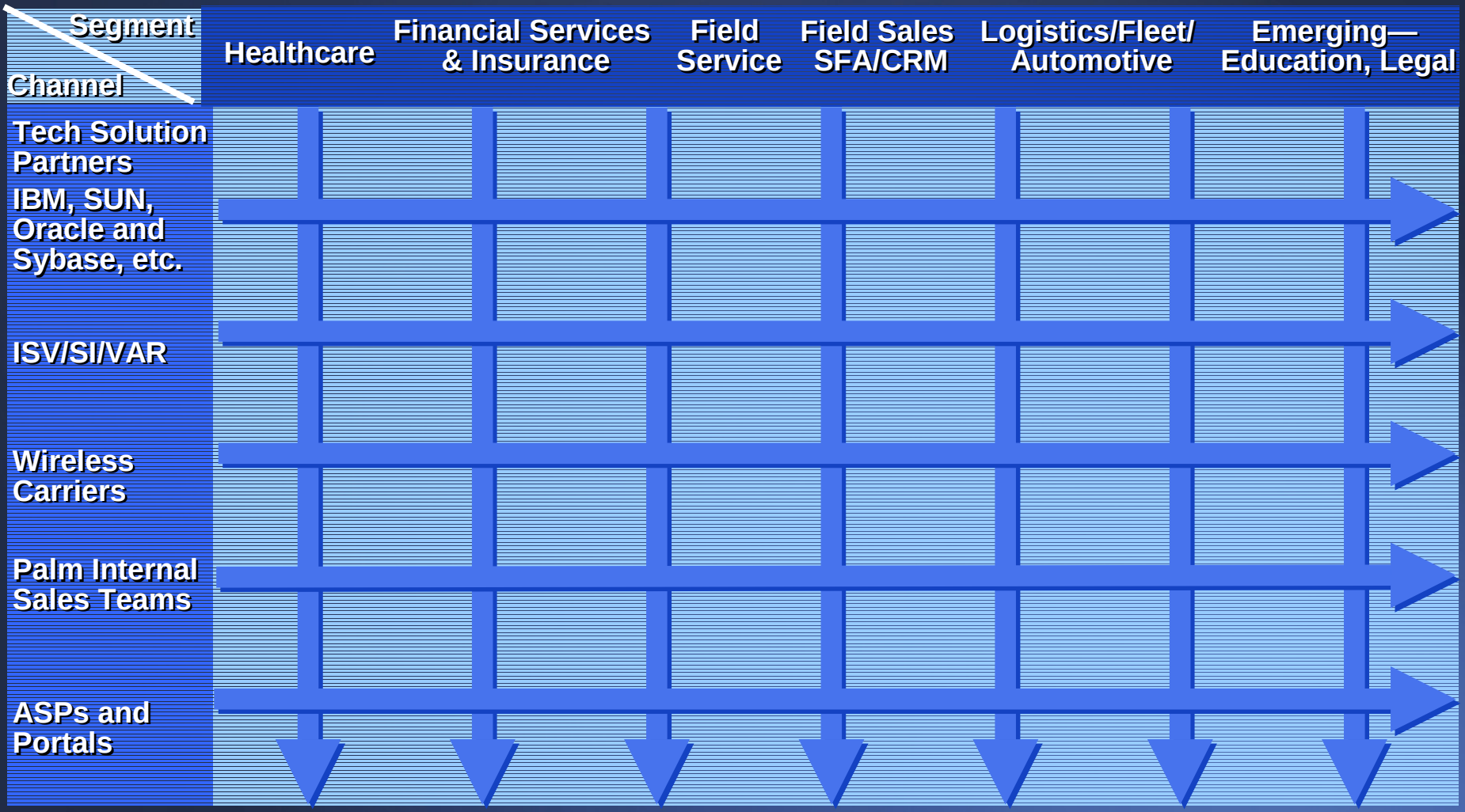


Source: Morgan Stanley Dean Witter Equity Research, IDC, The Yankee Group

Wireless Enterprise and Access



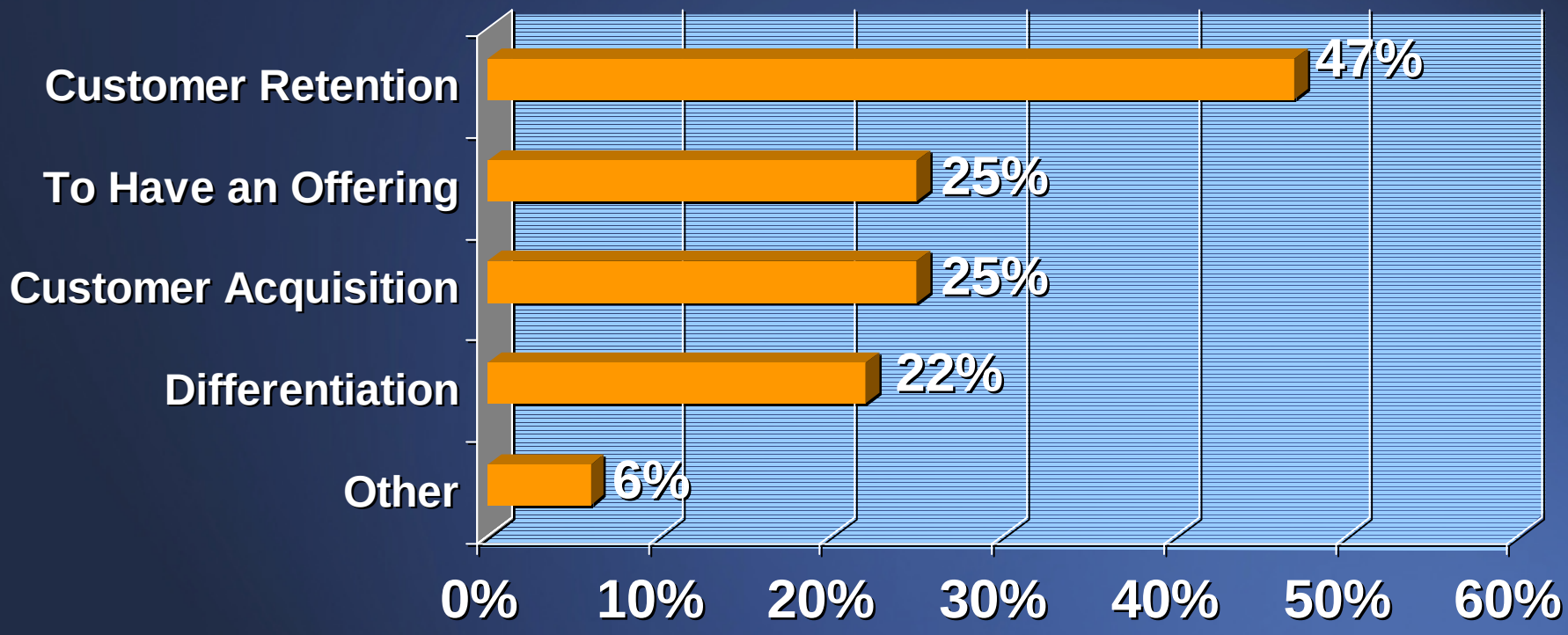
Target Wireless Enterprise Market Segments



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Financial Services: What Is Driving Mobile Commerce?

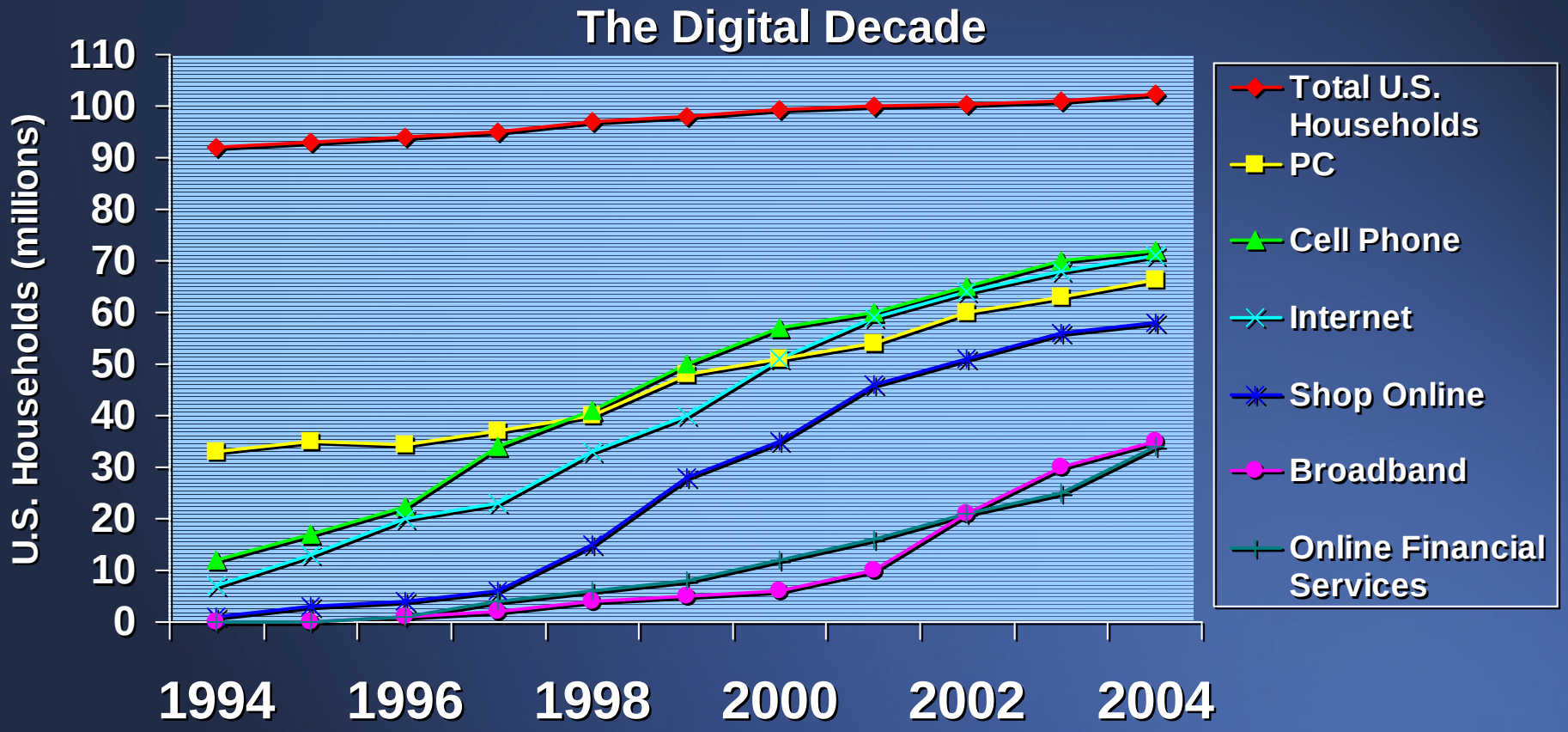
"What Benefits Do You Expect to Gain From a Wireless Strategy?"



Source: Forrester Research Inc.

Financial Services: What Is Driving Mobile Commerce?

Further Evidence of the Proliferation and Online Penetration



Source: Forrester Research Inc.

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Financial Services/MCommerce

Q: What Are the Opportunities?

A: Unlimited, Except by Imagination

- Retail Banking
- Mortgage Finance
- Consumer Finance
- Auto Finance
- Stock Brokerage
- Insurance
- Real Estate

Blue Chip Partner/Customer Base

- Banking: Sybase, 724, Bank of America, Wells Fargo, Chase, Citi
- Brokerage: Fidelity, ETrade, Ameritrade, DLJ, Charles Schwab, Merrill, Wtrade
- Payments/mCommerce: Verifone, LinkPoint, FDC
- Insurance: MDSI, Mitchell, DWL, Siebel
- And others…



Palm and LinkPoint



M-Commerce Rapid Application Development

- LinkPoint International is offering the LinkPoint Codewarrior™ Wrapper that empowers developers to create secure Palm™ applications for mobile commerce
  - Simple, well documented API
  - Handles LinkPoint Secure Payment Gateway protocols
  - Easily integrates into current Palm™ applications
  - Serves as a great foundation for new Palm™ applications
  - The Data Structure supports multiple credit card transactions:
    - Sale
    - Authorize Only
    - Return
    - Force Ticket
    - Ticket Only

Palm and LinkPoint Wireless PDA7

- New Software Application that Combines:
  - Efficiency
  - Reliability
  - Security
  - Mobility of a PDA



“Enabling Merchants to Process Credit Card Transactions Via the Internet Anytime, Anyplace”

Palm and iAnywhere Solutions

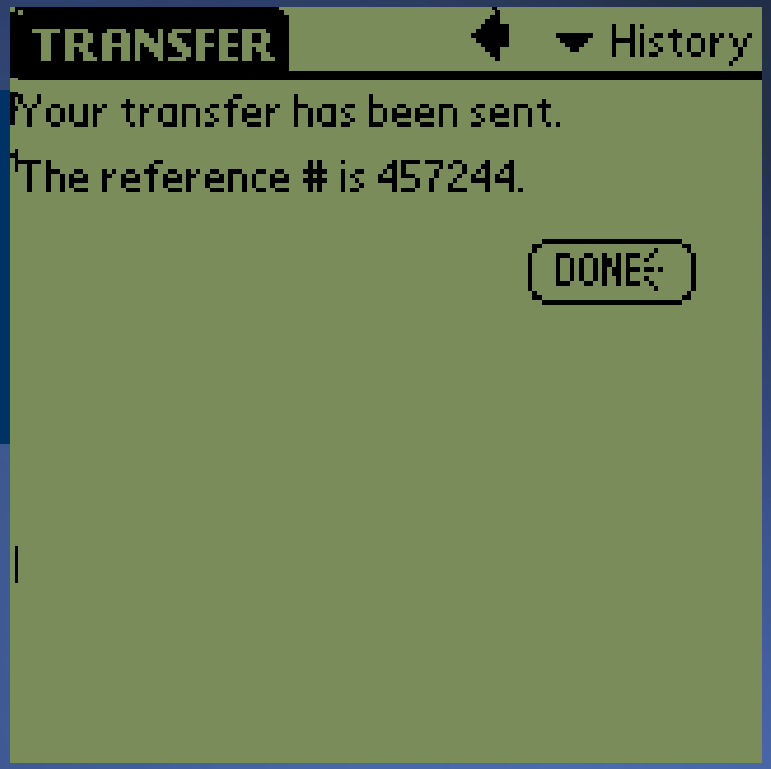


- Delivering Wireless e-Finance Solutions
  - Enables consumer banking and trading
  - Personalization of the user's experience; notification and alerts based on the users preferences and priorities
- Uniquely offer 'Always Available' Access
  - Users can work online and offline
  - Ability to queue outbound requests and asynchronously receive outbound messages
- Comprehensive Security
  - User authentication, authorization and wireless encryption

Palm and iAnywhere Solutions Example: Palm VII

Transfer Completed Screen

This screen confirms the transfer has been sent and displays the transaction reference number.



Summary: What Should I Do?

Do Something and Do Something Quickly!

- Have a solid Wireless Internet strategy
  - When ready, develop a strategic wireless plan for growth, with well-thought out infrastructure and offerings
  - Avoid temptation to jump in tactically, just to occupy space
- Select a vendor carefully to:
  - Conduct assessment up front
  - Determine functional suitability for wireless
  - Perform design work for high-potential functions to port properly to wide range of wireless devices
  - Analyze data to be ported and package to make content-rich

Summary: Palm's Focus on Financial Services

- Market Development
  - Led by Erika Mobley
  - 408-326-2462

- Business Development
  - Bill Faresich
  - mCommerce, Payment Processing & Insurance
  - 973-659-9919

  - Rob Sweeney
  - Retail Banking, Brokerage & Emerging
  - 816-943-8733

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